

# 2 Data Import and Wrangling

Mohan Khanal

2025-07-30

## Table of contents

|          |                                                                                     |          |
|----------|-------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Chapter 2: Data Import and Data Wrangling</b>                                    | <b>1</b> |
| 1.1      | Agenda . . . . .                                                                    | 1        |
| 1.2      | Part 1: Reading Different Types of Data (30 mins) . . . . .                         | 2        |
| 1.2.1    | CSV File Example: Car Dataset . . . . .                                             | 2        |
| 1.3      | Part 2: Data Cleaning and Manipulation using <code>dplyr</code> (40 mins) . . . . . | 2        |
| 1.3.1    | Load Required Packages . . . . .                                                    | 2        |
| 1.3.2    | Basic Wrangling Examples . . . . .                                                  | 3        |
| 1.4      | Part 3: Handling Missing Data (30 mins) . . . . .                                   | 4        |
| 1.4.1    | Checking Missing Values . . . . .                                                   | 4        |
| 1.4.2    | Removing or Replacing NAs . . . . .                                                 | 4        |
| 1.5      | Part 4: Practice and Q&A (20 mins) . . . . .                                        | 5        |
| 1.5.1    | Practice Tasks (based on car data) . . . . .                                        | 5        |
| 1.6      | Further Resources (for Data Import & Wrangling) . . . . .                           | 6        |

## 1 Chapter 2: Data Import and Data Wrangling

### 1.1 Agenda

- **Part 1 (30 mins)** – Data Import and Data Wrangling (Reading Different Types of Data)
- **Part 2 (40 mins)** – Data Cleaning and Manipulation using `dplyr`
- **Part 3 (30 mins)** – Handling Missing Data
- **Part 4 (20 mins)** – Practice and Q&A

## 1.2 Part 1: Reading Different Types of Data (30 mins)

We can also check our working directory using the command `getwd()`.

### 1.2.1 CSV File Example: Car Dataset

```
#we can import csv file using the base library using the command read.csv
car_data <-read.csv("../data/car_data.csv")

# or we can use readr library using the command read_csv to import csv

library(readr)
rcar_data <- read_csv("../data/car_data.csv")
```

Rows: 301 Columns: 9

-- Column specification -----

Delimiter: ","

chr (4): Car\_Name, Fuel\_Type, Seller\_Type, Transmission

dbl (5): Year, Selling\_Price, Present\_Price, Kms\_Driven, Owner

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
names(car_data)
```

```
[1] "Car_Name"      "Year"          "Selling_Price" "Present_Price"
[5] "Kms_Driven"    "Fuel_Type"     "Seller_Type"   "Transmission"
[9] "Owner"
```

## 1.3 Part 2: Data Cleaning and Manipulation using dplyr (40 mins)

### 1.3.1 Load Required Packages

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

## 1.3.2 Basic Wrangling Examples

### 1.3.2.1 Selecting and Filtering

```
# Select relevant columns and filter cars after 2015
car_data_filtered <- car_data %>%
  select(Car_Name, Year, Selling_Price, Fuel_Type) %>%
  filter(Year > 2015)
```

### 1.3.2.2 Creating New Variables with mutate()

```
# Calculate car age and classify it
car_data <- car_data %>%
  mutate(
    Car_Age = 2025 - Year,
    Age_Group = ifelse(Car_Age <= 5, "New", "Old")
  )
```

### 1.3.2.3 Sorting Rows with arrange()

```
# Sort by Selling Price
car_data_sorted <- car_data %>% arrange(desc(Selling_Price))
```

### 1.3.2.4 Summarizing with group\_by() and summarize()

```
# Average selling price by fuel type
car_data %>%
  group_by(Fuel_Type) %>%
  summarize(avg_price = mean(Selling_Price, na.rm = TRUE))
```

```
# A tibble: 3 x 2
  Fuel_Type avg_price
  <chr>      <dbl>
1 CNG        3.1
2 Diesel    10.3
3 Petrol     3.26
```

---

## 1.4 Part 3: Handling Missing Data (30 mins)

### 1.4.1 Checking Missing Values

```
library(naniar)
miss_var_summary(car_data)
```

```
# A tibble: 11 x 3
  variable      n_miss pct_miss
  <chr>         <int>   <num>
1 Car_Name         0         0
2 Year             0         0
3 Selling_Price    0         0
4 Present_Price    0         0
5 Kms_Driven       0         0
6 Fuel_Type        0         0
7 Seller_Type      0         0
8 Transmission     0         0
9 Owner            0         0
10 Car_Age         0         0
11 Age_Group       0         0
```

### 1.4.2 Removing or Replacing NAs

```
# Remove rows with any missing values
car_data_clean <- na.omit(car_data)

# Replace missing Selling_Price with 0 (if any)
car_data <- car_data %>%
```

```
mutate(Selling_Price = ifelse(is.na(Selling_Price), 0, Selling_Price))

head(car_data)
```

|   | Car_Name      | Year | Selling_Price | Present_Price | Kms_Driven | Fuel_Type |
|---|---------------|------|---------------|---------------|------------|-----------|
| 1 | ritz          | 2014 | 3.35          | 5.59          | 27000      | Petrol    |
| 2 | sx4           | 2013 | 4.75          | 9.54          | 43000      | Diesel    |
| 3 | ciaz          | 2017 | 7.25          | 9.85          | 6900       | Petrol    |
| 4 | wagon r       | 2011 | 2.85          | 4.15          | 5200       | Petrol    |
| 5 | swift         | 2014 | 4.60          | 6.87          | 42450      | Diesel    |
| 6 | vitara brezza | 2018 | 9.25          | 9.83          | 2071       | Diesel    |

  

|   | Seller_Type | Transmission | Owner | Car_Age | Age_Group |
|---|-------------|--------------|-------|---------|-----------|
| 1 | Dealer      | Manual       | 0     | 11      | Old       |
| 2 | Dealer      | Manual       | 0     | 12      | Old       |
| 3 | Dealer      | Manual       | 0     | 8       | Old       |
| 4 | Dealer      | Manual       | 0     | 14      | Old       |
| 5 | Dealer      | Manual       | 0     | 11      | Old       |
| 6 | Dealer      | Manual       | 0     | 7       | Old       |

## 1.5 Part 4: Practice and Q&A (20 mins)

### 1.5.1 Practice Tasks (based on car data)

- Import the car dataset
- Select cars with Present\_Price > 10
- Create a variable to calculate depreciation = Present\_Price - Selling\_Price
- Group by Transmission type and calculate average depreciation

```
car_data %>%
  mutate(depreciation = Present_Price - Selling_Price) %>%
  group_by(Transmission) %>%
  summarize(avg_depreciation = mean(depreciation, na.rm = TRUE))
```

```
# A tibble: 2 x 2
  Transmission avg_depreciation
  <chr>         <dbl>
1 Automatic     5.90
2 Manual        2.52
```

---

## 1.6 Further Resources (for Data Import & Wrangling)

- Tidyverse Documentation: <https://www.tidyverse.org/packages/>
- Importing Data in R: <https://r4ds.hadley.nz/data-import.html>
- Data Transformation Cheatsheet: <https://posit.co/resources/cheatsheets/>